

AI Top Chef

1. Function Description

After the program starts, give the large model a command asking what delicious food can be made from the ingredients on the desktop. The program will check the ingredients on the desktop and reply with what delicious dishes can be made. Finally, it will grip the ingredients and place them to the left of the robotic arm.

2. Startup

Open the terminal and enter the following command:

```
ros2 launch largemodel largemodel_control.launch.py text_chat_mode:=True
```

Then open a second terminal and enter the following command:

```
ros2 run text_chat text_chat
```

Then in the text_chat terminal, enter the following content:

```
what delicious food can be made from the ingredients on the desktop?
```

Press Enter. For the voice version, wake up the voice module and directly say "What delicious food can be made from the ingredients on the desktop?" to the voice module. Wait for the large model to think and reply, as shown in the figure below:

```
user input: Check what delicious food can be made with the ingredients on the desktop
okay@, let me think for a moment.../[INFO] [1775546132.275216146] [text_chat_node]: Decision making AI planning:You can only select a maximum of two ingredients. Follow these steps, and o
nly one action function will be output each time:
1. Call 'seewhat()' to observe the environment and see what ingredients are on the desktop.
2. Reply with the two ingredients that can be used to make a dish and the name of the dish.
3. Call 'seewhat()' to observe the position of the first ingredient.
4. Call 'grasp_obj(x1, y1, x2, y2)' to pick up the first ingredient.
5. Call 'putdown()' to put down the first ingredient.
6. Call 'seewhat()' to observe the position of the second ingredient.
7. Call 'grasp_obj(x1, y1, x2, y2)' to pick up the second ingredient.
8. Call 'putdown()' to put down the second ingredient.
[INFO] [1775546135.656126725] [text_chat_node]: "action": ['seewhat()'], "response": Okay, I'm ready to check what delicious food we can make! Let me take a look at the ingredients on the de
sktop first.
[INFO] [1775546144.734863623] [text_chat_node]: "action": ['seewhat()'], "response": I see several ingredients on the desktop, including tomatoes, mushrooms, and cauliflower. Let's make a de
licious tomato and mushroom stir-fry! I'll start by picking up the first ingredient.
[INFO] [1775546152.874208525] [text_chat_node]: "action": ['grasp_obj(320, 130, 510, 280)'], "response": I've located the tomato on the desktop. Now, I'll pick it up.
[INFO] [1775546180.698949757] [text_chat_node]: "action": ['putdown()'], "response": I've picked up the tomato! Now, I'll put it down in a suitable place.
[INFO] [1775546191.618077974] [text_chat_node]: "action": ['seewhat()'], "response": The tomato is now placed down. Next, I'll locate the mushroom to pick it up.
[INFO] [1775546200.120166039] [text_chat_node]: "action": ['grasp_obj(100, 200, 180, 270)'], "response": I've found the mushroom on the desktop. Now, I'll pick it up.
[INFO] [1775546228.872197541] [text_chat_node]: "action": ['putdown()'], "response": I've successfully picked up the mushroom! Now, I'll put it down next to the tomato.
[INFO] [1775546241.049346456] [text_chat_node]: "action": ['finishtask()'], "response": I've placed both the tomato and mushroom down. The ingredients for the tomato and mushroom stir-fry ar
e ready! I've completed all the tasks.
```

The robotic arm will select two ingredients that can be used to make a delicious dish based on what it sees, grip these two ingredients, and place them to the left of the robotic arm.

3. Task Planning

1. Call `seewhat()` to observe the environment and check what ingredients are on the desktop;
2. Reply with what two ingredients can be used to make a dish and the dish name;
3. Call `seewhat()` to observe the position of the first ingredient;
4. Call `grasp_obj(x1, y1, x2, y2)` to grip the first ingredient;
5. Call `putdown()` to place down the first ingredient;
6. Call `seewhat()` to observe the position of the second ingredient;
7. Call `grasp_obj(x1, y1, x2, y2)` to grip the second ingredient;

8. Call `putdown()` to place down the second ingredient.